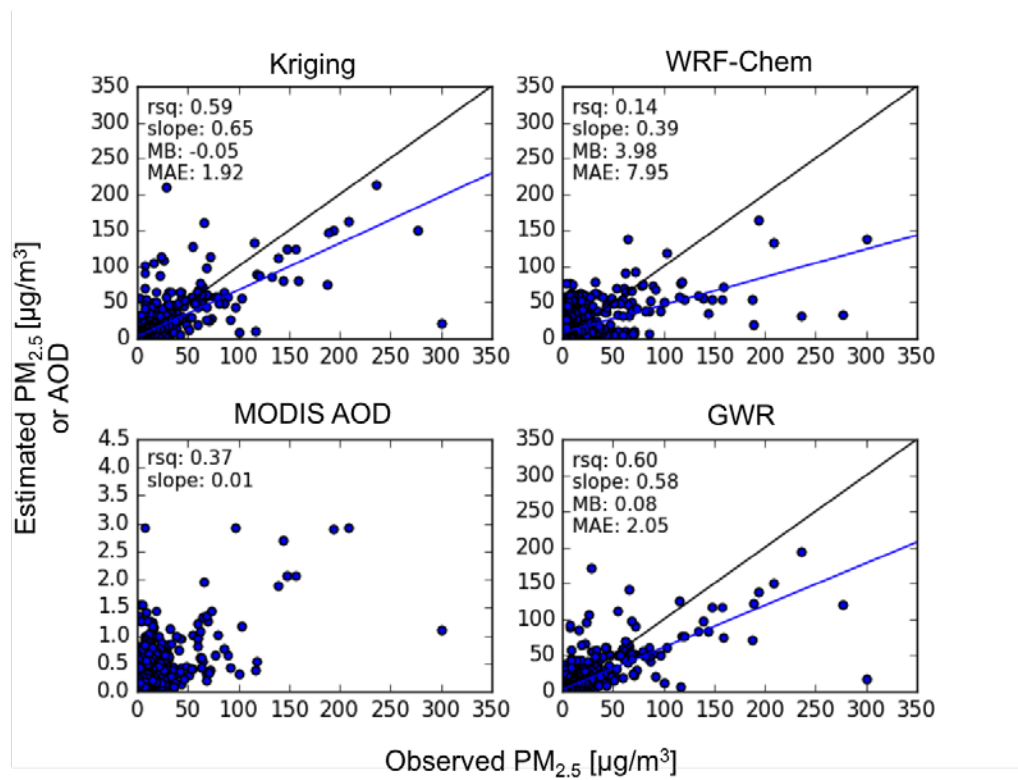




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595 **Figure S1:** Performance metrics of kriging surface-monitor interpolation model, Weather
596 Research Forecasting with Chemistry (WRF-Chem) chemical transport model, MODerate
597 Resolution Imaging Spectroradiometer (MODIS) aerosol optical depth (AOD), and
598 geographically weighted ridge regression (GWR) estimated $PM_{2.5}$ compared to $PM_{2.5}$ observed at
599 surface sites. Estimates from kriging and GWR are evaluated by leave-one-out cross validation
600 as the surface monitors are used to obtain these estimates.

Table S1: Same-day association between a 10 µg/m³ increase in WFS PM_{2.5} and risk for asthma-related care utilization event for all events (All Strata), adjusting for temperature.

Outcome	Strata	Odds Ratio	95% CI
Ambulance	All	1.259	0.752-2.107
Emergency Department	All	1.089	1.043-1.136
Inpatient Hospital	All	1.072	0.995 -1.154
Outpatient Hospital	All	1.065	1.029-1.103
Office	All	1.050	1.038-1.063
SABA Fill	All	1.077	1.065-1.088
Urgent Care Facility	All	0.809	0.552-1.187

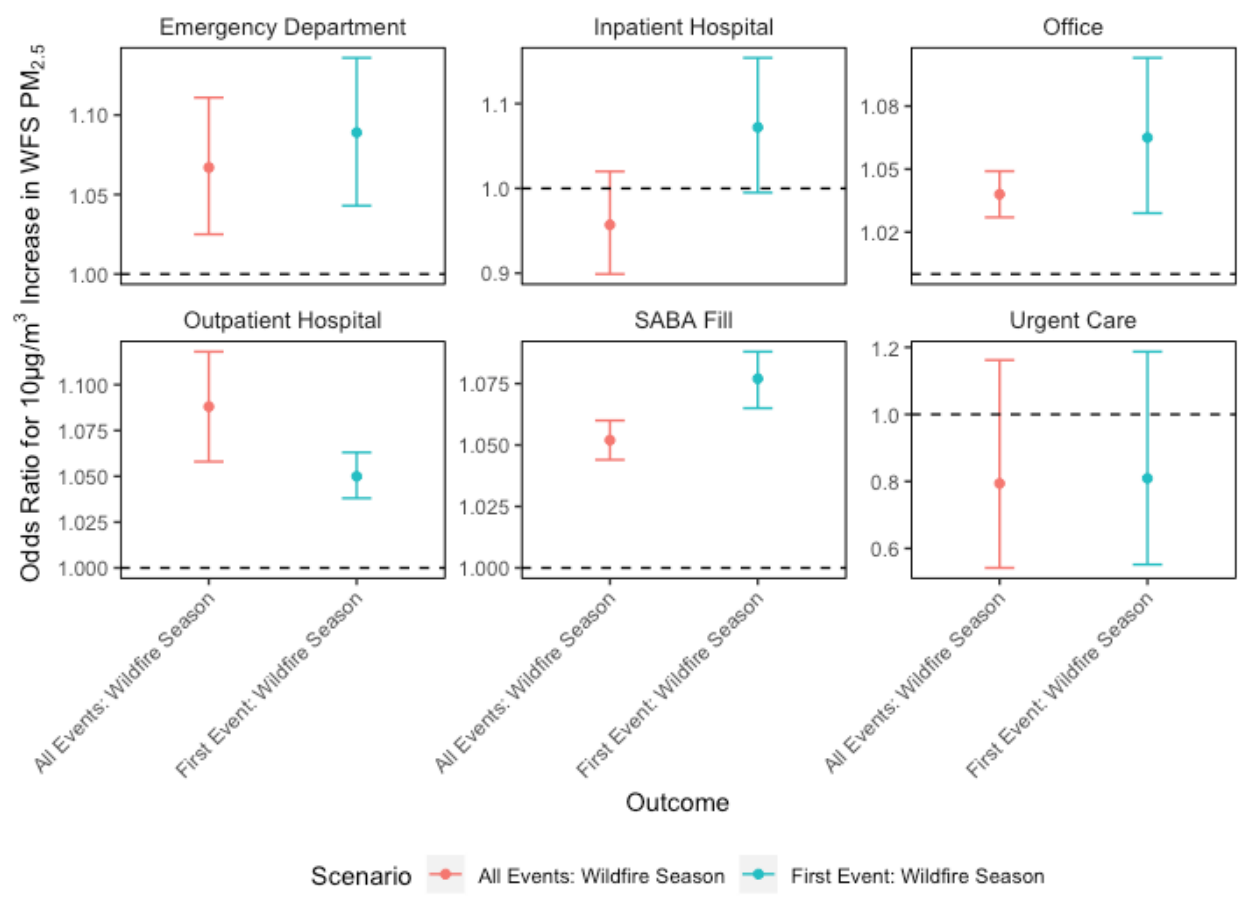
606 **Table S2:** Same-day association between a 10 $\mu\text{g}/\text{m}^3$ increase in WFS $\text{PM}_{2.5}$ and risk for asthma-
607 related care utilization event by sex, adjusting for temperature.

Outcome	Strata	Odds Ratio	95% CI
Ambulance	Male	--	--
Ambulance	Female	--	--
Emergency Department	Male	1.078	0.995-1.168
Emergency Department	Female	1.092	1.039-1.148
Inpatient Hospital	Male	1.113	0.955-1.297
Inpatient Hospital	Female	1.060	0.973-1.155
Outpatient Hospital	Male	1.079	1.022-1.140
Outpatient Hospital	Female	1.057	1.010-1.106
Office	Male	1.031	1.010-1.052
Office	Female	1.061	1.046-1.077
SABA Fill	Male	1.059	1.039-1.079
SABA Fill	Female	1.087	1.073-1.101
Urgent Care Facility	Male	0.716	0.345-1.487
Urgent Care Facility	Female	0.857	0.560-1.311

609 **Table S3:** Same-day association between a 10 $\mu\text{g}/\text{m}^3$ increase in WFS $\text{PM}_{2.5}$ and risk for asthma-
610 related care utilization event by age category, adjusting for temperature.

Outcome	Strata	Odds Ratio	95% CI
Ambulance	Age < 15	--	--
Ambulance	Age 15-65	--	--
Ambulance	Age > 65	--	--
Emergency Department	Age < 15	1.050	0.946-1.166
Emergency Department	Age 15-65	1.098	1.043-1.156
Emergency Department	Age > 65	1.096	0.977-1.229
Inpatient Hospital	Age < 15	1.073	0.891-1.292
Inpatient Hospital	Age 15-65	1.059	0.942-1.191
Inpatient Hospital	Age > 65	1.083	0.968-1.211
Outpatient Hospital	Age < 15	1.045	0.959-1.139
Outpatient Hospital	Age 15-65	1.076	1.030-1.123
Outpatient Hospital	Age > 65	1.049	0.965-1.140
Office	Age < 15	1.032	1.006-1.058
Office	Age 15-65	1.062	1.046-1.078
Office	Age > 65	1.036	1.005-1.068
SABA Fill	Age < 15	1.056	1.031-1.081
SABA Fill	Age 15-65	1.088	1.074-1.103
SABA Fill	Age > 65	1.056	1.024-1.090
Urgent Care Facility	Age < 15	1.064	0.578-1.958
Urgent Care Facility	Age 15-65	0.690	0.410-1.162
Urgent Care Facility	Age > 65	--	--

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Figure S2: Sensitivity analysis results comparing association between wildfire smoke (WFS)

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PM_{2.5} and asthma health care utilization (AHCU) using all events observed per person and first

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event observed per person (results in Figure 2) during the 2013-05-01 to 2013-09-30 wildfire

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season.

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Table S4: Same-day association between binary WFS PM_{2.5} cutoff and relative rate ratio for asthma-related care utilization event adjusting for temperature, seasonality, and weekend.

Outcome	WFS Cutoff	Rate Ratio	Rate Ratio 95% CI	Percent Attributed	Percent Attributed 95% CI
Emergency Department	>5 µg/m ³	0.986	0.617-1.577	-1.4	-62.1-36.6
Emergency Department	>10 µg/m ³	0.916	0.572-1.468	-9.2	-74.8-31.9
Emergency Department	>15 µg/m ³	1.103	0.679-1.792	9.3	-47.3-44.2
Inpatient Hospital	>5 µg/m ³	0.930	0.693-1.248	-7.5	-44.3-19.9
Inpatient Hospital	>10 µg/m ³	0.977	0.731- 1.361	-0.3	-36.8-26.5
Inpatient Hospital	>15 µg/m ³	1.063	0.766-1.475	5.9	-30.5-32.2
Outpatient Hospital	>5 µg/m ³	0.812	0.573-1.148	-23.2	-74.5-12.9
Outpatient Hospital	>10 µg/m ³	0.881	0.604-1.286	-13.5	-65.6-22.2
Outpatient Hospital	>15 µg/m ³	0.838	0.565-1.243	-19.3	-77.0-19.5
Office	>5 µg/m ³	1.062	0.873-1.292	5.8	-14.5-22.6
Office	>10 µg/m ³	1.318	1.060-1.639	24.1	5.7-39.0
Office	>15 µg/m ³	1.366	1.107-1.687	26.8	9.7-40.7
SABA Fill	>5 µg/m ³	1.157	0.983-1.363	13.6	-1.7-26.6
SABA Fill	>10 µg/m ³	1.311	1.097-1.566	23.7	8.8-36.1
SABA Fill	>15 µg/m ³	1.425	1.203-1.687	29.8	16.9-40.7
Urgent Care Facility	>5 µg/m ³	1.730	0.904-3.307	42.2	-10.6-69.8
Urgent Care Facility	>10 µg/m ³	1.730	0.904-3.307	42.2	-10.6-69.8
Urgent Care Facility	>15 µg/m ³	1.602	0.996-2.576	37.6	-0.4-61.2